GRADE 1 · BLOCK CODING

Forward and Back

Forward adds, back takes away

Name: .... Date: ....

★ **I can run a code that mixes forward and back blocks.**

Two kinds of moves

A forward block moves Bit UP the path (it adds steps). A back block moves Bit DOWN the path (it takes steps away). A code can use both. Run the blocks in order, one at a time, to find where Bit lands.

Bit's number path

0	1	2	3	4	5

Code A

start at 1

forward 3

back 1

1 Run Code A

Start on 1. Forward 3, then back 1. Run it one block at a time. What number does Bit land on?

Code B

start at 3

forward 1

back 2

2 Predict, then run Code B

PREDICT where Bit will land. Then run Code B from 3: forward 1, then back 2. What number does Bit land on?

Build C — start at 2, reach 4

start at 2

Use ONE forward block and ONE back block.

3 You try — mix the moves

Fill the 2 empty blocks with one forward and one back block so Bit goes from 2 to 4. Run it to check. What did you use?

Big idea

Forward adds steps and back takes steps away — just like adding and subtracting. A code can mix forward and back blocks to reach a number.

TEACHER GUIDE · GRADE 1 · BLOCK CODING

Forward and Back

Quick facilitation notes & answer key

What this worksheet teaches

Students run instructions that mix forward (add) and back (subtract) blocks, then write one of their own. Forward is like adding and back is like taking away — so coding the number line is doing math. No coding experience needed.

Before you start (no computer needed)

Paper and pencil only. Optional: a floor number line 0 to 5 so students can step forward and back to feel adding and taking away.

How to lead it (about 20 minutes)

1. Show them first (you do). Run Code A: "start at 1… forward 3 to 4… back 1 to 3." Name it: forward adds, back takes away.
2. Do one together (we do). Exercise 1: Code A lands on 3. Exercise 2: predict Code B, then run — forward 1 to 4, back 2 to 2.
3. Let them try alone (you do). Exercise 3: from 2, use one forward and one back block to reach 4 (e.g. forward 3, back 1).
4. Big idea: "Forward is like adding and back is like subtracting — coding the number line is doing math."

Answer key (these are the verified correct answers)

Exercise 1 — Code A: 1 → forward 3 → 4 → back 1 → 3. Bit lands on 3.

Exercise 2 — Code B: 3 → forward 1 → 4 → back 2 → 2. Bit lands on 2.

Exercise 3 — start 2 → 4: e.g. forward 3, back 1 (a net of forward 2). Any forward/back pair that nets +2 works.

Why it works: each block runs in order; forward adds its number and back subtracts its number, so the landing is start + forwards – backs (kept on the 0–5 line).

If students get stuck — and ways to adjust

Watch for: treating a back block as forward. Point to the back word and step the wrong way once to contrast.

Make it easier: do Codes A and B only; skip the build task.

Make it harder: ask for a 2-to-4 answer that uses a back block first (e.g. back 1 to 1, then forward 3 to 4).

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

In plain terms: students follow (or write) step-by-step instructions, in order, to get a result.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

In plain terms: students read instructions someone wrote, change one, and explain what the change did.

You'll know it worked when

The student lands Code A on 3 and Code B on 2 and writes a forward+back answer that reaches 4.